Chapter 2: Lines and Angles

Worksheet 2: Types of Lines and Angles

Learning Objective: Students will identify and classify lines (parallel, perpendicular, intersecting) and angles (acute, obtuse, right, straight, reflex); measure angles and find unknown angles using angle relationships.

Worked Example

Two lines intersect. One angle formed is 65. Find all four angles.
Solution: Vertically opposite angles are equal.
Adjacent angles are supplementary (add to 180).
Angles: 65, 115, 65, 115.

Questions 1–5: Easy / Recall

Q1. Define parallel lines. Give one example from daily life.

Q2. What is the measure of a right angle?

Q3. Classify the angle 135.

Q4. Two angles are supplementary. One is 70. Find the other.

Q5. Two angles are complementary. One is 42. Find the other.

Questions 6–10: Basic Application

Q6. Lines AB and CD are parallel. A transversal crosses them. One alternate interior angle is 55. What is the other alternate interior angle?

Q7. Two lines intersect at a point. One angle is 120. Find the other three angles.

Q8. How many degrees does a minute hand of a clock rotate in 15 minutes?

Q9. Name the type of angle formed by the hands of a clock at 9:00.

Q10. A straight road crosses another road. One angle formed is 75. What are all four angles at the intersection?

Questions 11–15: Practice and Skill Building

Q11. Find angle x: Three angles on a straight line are x, $2x$, and $3x$. Find x.

Q12. A triangle has angles x, $2x$, and $3x$. Find all three angles. What type of triangle is it?

Q13. In a quadrilateral, three angles are 90, 85, and 110. Find the fourth angle.

Q14. Co-interior (same-side interior) angles between parallel lines add to _____. If one is 65, find the other.

Q15. The angle of a clock at 4:00 is _____ degrees. Is it acute, right, or obtuse?

Questions 16–18: Higher Order Thinking

Q16. Lines l and m are parallel. A transversal cuts them. Corresponding angles are $3x + 10$ and $5x - 30$. Find x and the angle.

Q17. The angles of a triangle are in ratio $2 : 3 : 4$. Find each angle. Is any angle greater than 90?

Q18. Two parallel lines are cut by two transversals, forming a triangle. Two angles of the triangle are 50 and 70. Find the third angle. What type of triangle is formed?

Questions 19–20: Real-Life Applications

Q19. A ladder leans against a wall making an angle of 70 with the ground. What angle does it make with the wall? What type is each angle?

Q20. Two straight roads cross at a market. One angle at the crossing is 55. A traffic island (triangle) is built in one of the corners of the crossing with angles 55 and 90 inside it. Find the third angle of the triangle.

Reflection Question: Explain the difference between alternate angles and corresponding angles when a transversal crosses parallel lines. Draw a diagram to support your answer. (Teacher Verification Required)